Harshal Gadekar

83080 52795 | harshalgadekar72@gmail.com | linkedin.com/in/harshal-gadekar

Education

Rajarshi Shahu College of Engineering | CGPA – 9.1 Pune, India
Bachelor of Technology in Automation and Robotics, Minor in Computer Science *August 2023 – Present*

MIT Junior College Pune, India
Higher Secondary Certificate in Science *July 2021 – March 2023*

Experience

Founder, Rocketry and Research Club Dec 2025 – Present
Rajarshi Shahu College of Engineering *Pune, India*

- Led cross-functional teams to deliver complex engineering projects under strict deadlines.
- Managed project lifecycles and resource allocation for high-stakes aerospace initiatives.
- Enforced rigorous safety protocols and risk management for hazardous hardware testing.

Team Lead, Janatics Automation Skill Challenge (JASC) June 2025 – Nov 2025
Rajarshi Shahu College of Engineering *Pune, India*

- Optimized automated workflows to maximize throughput and eliminate manual bottlenecks.
- Oversaw procurement and supply requirements to ensure delivery within technical constraints.
- Improved system reliability and operational accuracy during high-pressure competitions.

Team Lead, Smart India Hackathon (SIH) June 2024 – Aug 2024
Rajarshi Shahu College of Engineering *Pune, India*

- Analyzed logistical bottlenecks to engineer resource management systems for high-volume environments.
- Translated complex operational needs into scalable and functional technical architectures.
- Supervised prototyping and testing to ensure solutions met performance and efficiency metrics.

Research work

Research paper | ***XAI & Cognitive Overload*** Nov 2025 – Present

- Initiated XAI framework to mitigate operator cognitive overload and improve safety.
- Analyzed human-factors to enhance transparency during human-robot handover protocols.
- Oversaw design optimization to ensure scalable and high-performance automated solutions.

Design Patent | *High-Tech Rover Adaptive Suspension* Jan 2026 – Present

- Led design lifecycle to optimize weight, mechanical reliability, and system durability.
- Directed modelling to ensure high-performance outcomes in extreme environments.
- Managed mechanism development to improve operational stability and hardware longevity.

Technical Skills

Operations: *Process Optimization, Workflow Automation, Risk Management, Resource Utilization*
Design: *SolidWorks, Fusion360, 3D printing*
Simulation: *MATLAB/Simulink, robot kinematics and dynamics*
Programming: *Python, C/C++, ROS*

Certifications

Foundations of Project Management - *Google*
Design Thinking for Innovation – *University of Virginia*
Rocket Propulsion and Spacecraft Dynamics - *Kodacy*
Certified Entry Level Python Programmer - *Udemy*

Achievements

Janatics Automation Skill Challenge *(semi-finals)*
Collaborated with Purdue University *(EPICS)*
Smart India Hackathon *(institute level)*